GATE CS & IT: 10-Year Subject-Wise Question Bank & Solutions

Subject: Competitive Examination Revision | Author: Rohan Deshmukh (BITS Pilani (GATE AIR 42)) | 100% Free Campus Access

1. THEORY OF COMPUTATION (TOC)

• **Chomsky Hierarchy:** Regular (Type 3 / DFA/NFA) subset of Context-Free (Type 2 / Pushdown Automata) subset of Context-Sensitive (Type 1 / LBA) subset of Recursively Enumerable (Type 0 / Turing Machine).
• **Pumping Lemma for Regular Languages:** If L is regular, there exists pumping length p such that any string s with |s| >= p can be split into s = xyz where |xy| <= p, |y| > 0, and xy^i z in L for all i >= 0. Common counterexample: a^n b^n.

2. OPERATING SYSTEMS & COMPUTER ARCHITECTURE

• **Virtual Memory Paging:** Physical Address = Frame Number * Frame Size + Offset. Effective Memory Access Time (EMAT) = Hit_TLB * (TLB_access + RAM_access) + Miss_TLB * (TLB_access + 2 * RAM_access).
• **Pipeline Hazards:** Speedup = (k * n) / (k + n - 1), where k = number of pipeline stages, n = number of instructions. Structural, Data (RAW, WAR, WAW), and Control hazards.

3. COMPUTER NETWORKS & TCP/IP

• **TCP Flow & Congestion Control:** Slow start doubles congestion window (CWND) every RTT until SSThresh is reached. Additive Increase Multiplicative Decrease (AIMD) increments CWND by 1 MSS per RTT upon successful ACK.
• **Subnetting & CIDR:** Given IP 192.168.1.0/26: Number of subnets = 4, Total addresses per subnet = 64 (2^(32-26)), Usable host addresses = 62 (excluding network ID and broadcast address).